Christopher Ling

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Education

Georgia Institute of Technology Atlanta, GA

B.S. in Computer Science; GPA: 3.42/4.00 *Expected December 2026*

- Concentrations: Systems & Architecture; Information Internetworks
- Relevant coursework: Data Structures & Algorithms, Software Engineering, Database Systems, Computer Networking, Advanced Networking, Operating Systems, Advanced Computer Architecture, Probability & Statistics, Agile/Scrum, Prompt Engineering

Technical Skills

Languages: Python, Java, JavaScript/TypeScript, C, SQL (MySQL), HTML/CSS
Frameworks & Libraries: React.js, React Native, Next.js, Spring Boot, Flask, Tailwind CSS
Cloud & Backend: Google Cloud Platform (GCP), Firebase, Appwrite
Tools & Developer Environments: Git, Docker, Jira, OpenAI API, Android Studio, IntelliJ IDEA

Projects

Secure Mobile Messaging Platform | *React Native, TypeScript, Appwrite* Aug. 2025 – Dec. 2025

- Integrated an Appwrite backend to manage real-time data synchronization, secure user authentication, and persistent message storage.
- Enforced strict type safety and optimized state management using TypeScript, reducing runtime errors and improving codebase maintainability.
- Designed a responsive, mobile-first UI using React Native, ensuring a seamless cross-platform experience for iOS and Android users.

Stock News Predictor | *Next.js, Python, Firebase, OpenAI API, GCP* May 2025 – Jul. 2025

- Built a Python data pipeline to aggregate and process real-time financial news, mapping market sentiment to specific stock equity movements.
- Optimized API calls to OpenAI using advanced prompt engineering techniques to ensure context-aware and financially accurate chatbot responses.
- Implemented secure Firebase Authentication supporting Google and email/password sign-in
- Designed a responsive Next.js frontend to display dynamic stock trends and sentiment analysis metrics alongside the chat interface.

Flight Tracking Database System | *Python, Flask, MySQL, HTML/CSS* Jan. 2025 – May 2025

- Modeled complex relational database schemas utilizing foreign keys, indexing, and constraints to ensure ACID compliance for simulated flight operations.
- Built a responsive frontend dashboard using HTML/CSS to visualize flight routes and passenger manifests, seamlessly connected to the backend.
- Implemented error handling and input validation on Flask REST API endpoints to prevent SQL injection and ensure data integrity.